Flexible Gabor-Based Superpixel-Level Unsupervised LDA for Hyperspectral Image Classification

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***Abstract*— Hyperspectral images encompass abundant information and provide unique characteristics for material classification. However, the labeling of training samples can be challenging in hyperspectral image classification. To address this problem, this study proposes a framework named flexible Gabor-based superpixel-level unsupervised linear discriminant analysis (FG-*Su*ULDA) to extract the most informative and discriminating features for classification. First, a number of 3-D flexible Gabor filters are rigorously designed using an asymmetric sinusoidal wave to sufficiently characterize the spatial–spectral structure in hyperspectral images. Then, an unsupervised linear discriminant analysis strategy guided by the entropy rate superpixel (ERS) segmentation algorithm, called *Su*ULDA, is skillfully introduced to reduce the extracted large amount of FG features. The *Su*ULDA method not only boosts the classification capability but also increases the peculiarity of features, with the aid of superpixel information. Finally, the achieved features are imported to the popular support vector machine classifier. The proposed FG-*Su*ULDA framework is applied to four real hyperspectral image data sets, and the experiments constantly prove that our FG-*Su*ULDA is superior to several state-of-the-art methods in both classification performance and computational efficiency, especially with scarce training samples. The codes of this work are available at http://jiasen.tech/papers/ for the sake of reproducibility.**

***Index Terms*— Feature extraction, hyperspectral image classification, linear discriminant analysis (LDA), superpixel segmentation.**

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I. Introduction

HYPERSPECTRAL imagery (HSI) contains hundreds of spectral bands and offers opportunities for pixel-oriented material classification [1]. Commonly, a prerequisite step of applying classifiers is to manually label the hyperspectral pixel vectors in a scene [2]; however, this can be challenging and costly in certain applications. The relatively small size of labeling samples compared to the high dimension of the data would arise the curse of dimensionality (the so-called "Hughes phenomenon") [3]–[5]. Thus, the study on extracting distinguishable and important features for improving the classification performance with limited labeled samples has always been a hot topic. A variety of dimensionality reduction (DR) methods were proposed to reduce the number of features in the hyperspectral image. The DR task can be either fulfilled by feature selection (i.e., ranking-based methods and cluster-based methods), which removes redundant features and retains the important features in the original space [6]–[9], or feature extraction, which transforms the data from high dimensions to low ones by projecting to a new subspace [10], [11]. For example, the regularized linear discriminant analysis (RLDA) method regularizes the nonzero eigenvalues for feature reduction [12]. Fauvel *et al.* [13] proposed the kernel principal component analysis (KPCA) that uses a Gaussian kernel to implement nonlinear subspace projection and feature reduction. The spatial–spectral Schroedinger eigenmaps (SSSE) algorithm optimized the classifier by utilizing nondiagonal potentials in the Schroedinger eigenmap framework and integrating spatial and spectral information [14]. Nevertheless, most methods focused on spectral features only and ignored spatial features, leading to poor classification performance.

More recently, many feature extraction methods investigate spatial contextual information of ground materials [15]–[18]. Examples include filtering methods, such as morphological attribute profile, low-rank representation, and joint sparse representation, which explore the spatial regularity of hyperspectral data [19]–[22]. Intrinsic image decomposition and multiscale total variation methods can retain effective spatial feature [23], [24]. The edge-preserving operator is

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also combined with multiple probability maps [25] or superpixel spectral properties [26]. Local binary patterns (LBPs) model can gather local spatial statistical and semantic information [27]. However, these approaches process spatial and spectral information separately; the inherent 3-D correlations of the hyperspectral image cannot be fully characterized, resulting in low efficiency of representing the 3-D structure of the hyperspectral image.

Therefore, the spatial–spectral information in the hyperspectral image should be simultaneously considered by 3-D operators. Specifically, 3-D operators treat hyperspectral image data as a cube instead of a matrix or a vector. The 3-D discrete wavelet transform (3-D-DWT) filter is applied to hyperspectral data in three directions, respectively, which could capture the geometrical and statistical spatial–spectral information in hyperspectral data [28]. In [29], a 3-D morphological profile (3-D-MP) is introduced to exploit the dependence among the hyperspectral image. In particular, since Gabor wavelet is capable of achieving optimal spatial–spectral resolutions of signal [30], it has demonstrated powerful representative capability in hyperspectral image processing and drawn considerable attention in the literature [31]. Concretely, a series of 3-D Gabor filters with different scales and orientations are first defined. Through directly convolving with the original hyperspectral image, the obtained 3-D Gabor magnitude features can characterize the texture structure and statistical direction information in the local 3-D region [32]. Furthermore, a nonparametric and effective l_1-norm collaborative representation method is developed to decrease the computational cost and increase the discriminative ability [33]. Alternatively, a low-rank decomposition scheme is utilized for the Gabor feature extraction to accelerate the computation [34]. In fact, all these 3-D Gabor-based methods employ the classical circular 3-D Gabor filters, and thus, the structural information of various materials cannot be well described. Moreover, how to effectively take advantage of the huge number of extracted Gabor features is also a crucial issue for hyperspectral image classification.

In addition, anisotropic filtering methods have been introduced in the literature. Shearlet performs multiscale and multidirectional decomposition on the original data through shear matrix [35]. Curvelets, which is an improved version of Fourier transform, can effectively suppress the surrounding noise while obtaining multidirectional geometric details [36]. These multidirectional wavelets can perform a multiscale detailed analysis of data through a flexible transformation. Also, the anisotropic Gaussian filtering method can adjust the bandwidth factor and inclination according to the characteristics of the feature distribution [37], [38]. However, in existing research, the Gabor filtering is mostly used as an isotropic method, while nonhomogeneous information could be easily filtered out when processing hyperspectral image.

Many powerful deep learning-based methods for hyperspectral image processing are reviewed in [39]–[42]. The stacked autoencoder was the first trial to extract information for classification [43]. Subsequently, 1-D convolutional neural networks (CNNs), 2-D CNNs, and 3-D CNNs were then designed to carry out the feature extraction separately [44]–[46]. Regarding the hyperspectral image as sequential data, a recurrent neural network (RNN) was proposed to improve the classification performance [47], [48]. Furthermore, long short-term memory (LSTM) and CNNs are combined to extract spectral and spatial features [49]. Wu and Prasad [50] developed a semisupervised CNNs method for feature representation. Besides, several advanced frameworks, such as R-vertex component analysis network (VCANet) and MugNet, are introduced to address the limited training samples problem [51], [52]. However, most deep learning-based methods require adequate training samples to effectively estimate a large number of parameters in the optimization model (it is worth to point out that the training samples needed in [51] and [52] are still far beyond the practical situation), and thus, the practicability of deep learning-based methods can be deteriorated.

In this article, we aim to address the small sample set problem (the circumstances that the number of training samples of each class is very rare are mainly exploited) and strengthen the generalization of the proposed method as well. To this end, a flexible Gabor-based superpixel-level unsupervised linear discriminant analysis (FG-*Su*ULDA) framework is specially developed for hyperspectral image classification. First, through extending the sinusoidal plane wave in the 3-D Gabor filter into a more general form [53], a set of 3-D flexible Gabor wavelets is established to sufficiently examine the diverse spatial–spectral structural distribution characteristics of materials. Concretely, five types of a sinusoidal wave, including the circular 3-D Gabor filter, are carefully selected for feature extraction.

Second, the linear discriminant analysis (LDA) algorithm is introduced to remove the redundancy and retain the most discriminative content from the achieved a large amount of 3-D flexible Gabor (FG) features. Specifically, an unsupervised LDA strategy with the guidance of an entropy rate superpixel (ERS) segmentation method [54], named *Su*ULDA, is elaborated on to eliminate the dependence of traditional LDA algorithm on training samples, and the robustness of *Su*ULDA can be largely promoted. In particular, since it is reasonable to regard the pixels in a single superpixel as the same class with a high probability, the within-superpixel scatter matrix and between-superpixel scatter matrix of our *Su*ULDA are, respectively, computed and combined to complete feature reduction task in an unsupervised manner. Finally, the obtained features are concatenated together as input to the support vector machine (SVM) classifier. Fig. 1 shows the schematic of the proposed FG-*Su*ULDA approach to make the method more easily to be understood, and the widely used Salinas hyperspectral image data set is utilized. The novelty of this article mainly includes the following three aspects:

1) With respect to a 3-D Gabor filter, the traditional circular form has the ability to depict the spatial–spectral texture properties of objects, whereas the boundary regions (such as edges and blobs) of materials can be oversmoothed, and the corresponding discriminative ability can be weakened. Alternatively, the generalized sinusoidal plane wave in the proposed flexible Gabor wavelets is modified by introducing several parameters

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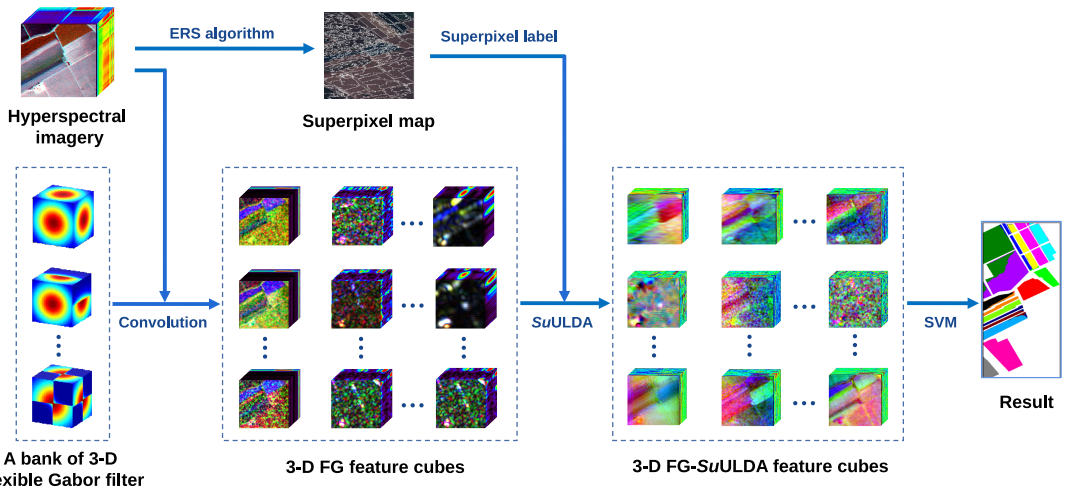


Fig. 1. System block diagram of the proposed FG-*Su*ULDA framework for hyperspectral image classification.

to control the degree of freedom, and the expressivity of capturing the various structural distribution of materials can be exactly improved, which is, thus, more consistent with the hyperspectral image. In our design, five groups of parameters are specially selected and utilized for Gabor feature extraction, so abundant informative features can be fulfilled.

2) The *Su*ULDA algorithm is designed as an unsupervised dimension reduction method with superpixel segmentation guidance, which is considered as the most important contribution of the proposed FG-*Su*ULDA framework. Traditionally, the LDA method compresses the feature data into a low-dimensional space according to the class separability of training samples, which is computed based on the relation between the so-called within- and between-class scatter matrices. Frankly speaking, LDA belongs to the supervised DR category, but LDA struggles with the scarce training sample problem, which has an essential influence on projection stability and accuracy. Fortunately, inspired by the homogenous attribute of superpixel in the segmentation map, each superpixel is considered as a single entity, and unsupervised LDA is subtly developed through conducting the peculiarity of superpixel-oriented scatter matrices. The proposed *Su*ULDA approach can not only improve the discriminative ability of the reduced features but also ensure the generalization ability of the whole DR process. Besides, our *Su*ULDA also erodes the inefficient performance of the 3-D Gabor feature on the abrupt change area of materials due to the incorporation of superpixel-level structural description.
3) Since the parameters of 3-D flexible Gabor filter can be preset and kept unchanged during feature extraction, while the well-designed *Su*ULDA method is totally unsupervised (it is worth to point out that the number of superpixels for different hyperspectral images can be heuristically determined), our FG-*Su*ULDA framework ensembles the two modules and can be carried out without manual intervention. Meanwhile, the whole FG-*Su*ULDA procedure is executed only once for each hyperspectral image due to irrelevance with the training set, while *Su*ULDA is sequentially applied on each module of the extracted FG feature cubes (rather than all FG features in a one-shot); therefore, the adaptability and efficiency of *Su*ULDA can be ensured, and it can be utilized for hyperspectral images with large coverage.

TABLE I
ABBREVIATIONS USED IN THIS ARTICLE

Abbreviation	Meaning
ERS	Entropy rate superpixel segmentation
3D-DWT	Three-dimensional discrete wavelet transform
3D-MP	Three-dimensional morphological profile
GCN	Graph convolutional network
GCK	Generalized composite kernel machine
KPCA	Kernel principal component analysis
OTVCA	Orthogonal total variation component analysis
NMFL	Nonlinear multiple feature learning
LBP-ELM	Local binary patterns and extreme learning machine
CGabor	Three-dimensional circular Gabor magnitude feature
RLDA	Regularized linear discriminant analysis
*Su*ULDA	Superpixel-level unsupervised LDA
FG-*Su*ULDA	Flexible Gabor-based *Su*ULDA

A number of state-of-the-art alternatives are considered in comparison, and four real hyperspectral image data sets are employed for evaluation. Experimental results consistently validate the superiority of the extracted features achieved by the proposed FG-*Su*ULDA framework, and the related classification performance is more advantageous when the number of training samples is very limited.

The rest of this article is organized as follows. Section II introduces some work related to our framework, including the 3-D Gabor wavelet, the ERS segmentation, and the LDA algorithm. Section III presents the proposed FG-*Su*ULDA framework. Experiments with four real hyperspectral image data sets are described in Section V. Finally, in Section VI, the conclusion is given. Besides, the primary acronyms that appear in the article are summarized in Table I.

II. RELATED WORKS

A. 3-D Gabor Wavelet

Mathematically, the Gabor wavelet is a sinusoidal plane wave modulated with a Gaussian kernel function.

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For a point (x, y, b) [(x, y) represents its spatial coordinate, and b represents spectral band] in the 3-D domain, the 3-D Gabor wavelet is circular and defined as follows [55]:

$$\Psi(x, y, b) = u(x, y, b) \times v(x, y, b) \tag{1}$$

where $u(x, y, b)$ is the Gaussian envelope and $v(x, y, b)$ is the sinusoidal plane wave function. $u(x, y, b)$ and $v(x, y, b)$ are defined, respectively, as

$$\begin{aligned} u(x, y, b) &= \frac{1}{s} \times \exp\left(-\frac{x^2 + y^2 + b^2}{2\sigma^2}\right) \\ v(x, y, b) &= \exp\left(j2\pi\left(F_x x + F_y y + F_b b\right)\right) \end{aligned} \tag{2}$$

where

$$\begin{aligned} s &= (2\pi)^{3/2}\sigma^3 \\ F_x &= f \sin\varphi \cos\theta \\ F_y &= f \sin\varphi \cos\theta \\ F_b &= f \cos\varphi \end{aligned} \tag{3}$$

where s is a normalized value, and σ is the width of Gaussian component in the spatial–spectral domain. (F_x, F_y, F_b), respectively, describes the attribute of each sinusoidal wave and can be comprised of three parameters: f is the central frequency of the sinusoidal component, while θ and φ are orientations of the wave vector in frequency domain. As the internal spatial–spectral structure of the hyperspectral image data is uncertain, different parameters of scales f and orientations (θ, φ) are preset to capture sufficient and distinguishing features for classification. Commonly, the orientation parameters change by steps of 45°, resulting in 13 Gabor filters for each frequency (the filter with $\varphi = 0$ corresponds to one wavelet rather than four). The specific values of parameters (f, θ, φ) are

$$\begin{aligned} f &\in \left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\right] \\ (\theta, \varphi) &\in \left[0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\right]. \end{aligned} \tag{4}$$

By changing parameters of scales and orientations, 52 Gabor filters are eventually gathered, and Z descriptors are mathematically denoted as $\Psi_z, z = 1, \ldots, Z$ $(Z = 52)$.

B. ERS Segmentation

ERS is a superpixel segmentation method that divides a hyperspectral image into K perceptually uniform regions, where K is the number of superpixels [54] (it is worth pointing out that the ERS method is presented based on the 3-D hyperspectral image rather than the common 2-D image to make this article more elegant). This method initializes an undirected graph $U = (P, D)$ to describe the hyperspectral image, where P represents the pixel set and D denotes the edge set. It aims to search a subset of edge set $A \subseteq D$ so that $U_E = (P, A)$ divides the image into K superpixels that are compact, homogeneous, and balanced. Specifically, the entropy rate of a random work on a graph $\mathcal{H}(A)$ and a balancing term on the cluster distribution $\mathcal{B}(A)$ are utilized. The objective function is defined as

$$\arg\max_{A} \quad \mathcal{H}(A) + \alpha\mathcal{B}(A) \quad \text{s.t. } A \subseteq D \tag{5}$$

where $\alpha > 0$ is the coefficient of balance between two terms, and the definitions of two terms are explained in [54].

Thus, K nonoverlapping superpixels are extracted from hyperspectral data. Pixels in the same superpixel are very likely labeled as the one type of material. Similar to an unsupervised algorithm that produces K clusters, the superpixel segmentation procedure allocates each pixel (x, y) to a single cluster.

Furthermore, the ERS method can well describe the spatial homogeneity of various materials and make the spatial distribution of the superpixel map relatively uniform. In the achieved superpixel segmentation map, the number of pixels in each superpixel is nearly consistent. The value of K is computed by a heuristic function as follows [56]:

$$K = \text{floor}\left(\frac{X \times Y}{1000 \times \delta^{\sqrt{\text{Res}}}}\right) \tag{6}$$

where floor means the rounding down operation to the nearest integer, $X \times Y$ is the spatial coverage of the hyperspectral image, and Res is the spatial resolution of the hyperspectral image. The parameter δ should be less than 1 and set as 0.7 in [56]. Since the values of all parameters in (6), including X, Y, δ, and Res, can be preset or known in advance, the computation of K, which is the most important parameter in superpixel segmentation method, can be automatically assigned.

C. Linear Discriminant Analysis

LDA is a feature reduction algorithm that transforms data into a new subspace [57]. LDA searches for a direction along which the distance between the same clusters is minimized and the distance between different clusters is maximized. Suppose that $\boldsymbol{a}$ denote the projection direction, and the objective function of LDA is defined as

$$\arg\max_{\boldsymbol{a}} \frac{\boldsymbol{a}^T \boldsymbol{C}_b \boldsymbol{a}}{\boldsymbol{a}^T \boldsymbol{C}_w \boldsymbol{a}} \tag{7}$$

where $\boldsymbol{C}_b$ is the between-class scatter matrix, and $\boldsymbol{C}_w$ is the within-class scatter matrix. The denominator describes the distance from the same class in the subspace where the numerator describes the distance from different classes. A considerable number of labeled samples are usually required for implementing LDA, which is obviously not the situation in hyperspectral image classification. Alternatively, by taking advantage of the homogenous attribute of the ERS segmentation map, this study uses the ERS algorithm to conduct the feature reduction process, and an unsupervised LDA version is elaborated.

III. Flexible Gabor-Based Superpixel-Level Unsupervised LDA Framework

A. 3-D Flexible Gabor Filter

As referred by the definition of the sinusoidal wave $v(x, y, b)$ in (2), the responses of circular 3-D Gabor feature

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are homogeneous in each direction, so only uniform information from adjacent spectral bands and spatial pixels can be granted. However, the internal spatial–spectral structure in hyperspectral image is actually inhomogeneous, due to the uneven spatial distribution and various spectral configuration. To be more clear, the circular 3-D Gabor filter in (1) is isotropic, which means that the neighboring information can only be indistinctively aggregated into the central point after convolution; nevertheless, the abrupt change (such as edges and blobs) of materials may be oversmoothed, and the correlated discriminative ability could be inevitably deteriorated. Fortunately, in order to explore different structural characteristics of materials and increase the expressivity of complex spatial–spectral correlations, a 3-D flexible sinusoidal wave $v_{\boldsymbol{\lambda}}(x, y, b)$ is designed as follows:

$$v_{\boldsymbol{\lambda}}(x, y, b) = \exp\left(j2\pi\left(F_x x^{\lambda_1} + F_y y^{\lambda_2} + F_b b^{\lambda_3}\right)^{\frac{1}{\lambda_4}}\right) \tag{8}$$

where $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ is a set of parameters that control the grating degree of the oriented complex sinusoidal wave. Concretely, the parameters (λ_1, λ_2) describe the response intensity in spatial domain, while λ_3 depicts the intensity in spectral domain. The last parameter λ_4 adjusts the response intensity of the whole structure and is calculated as

$$\lambda_4 = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}. \tag{9}$$

By importing the set of parameters $\boldsymbol{\lambda}$, the function for calculating the 3-D flexible Gabor wavelet [referred to (1) and (2)] becomes

$$\Psi_{\boldsymbol{\lambda}}(x, y, b) = u(x, y, b) \times v_{\boldsymbol{\lambda}}(x, y, b). \tag{10}$$

Similarly, by taking different values of scales f and orientations (θ, φ) in (4), $\Psi_{\boldsymbol{\lambda}}$ also produces $Z = 52$ 3-D flexible Gabor wavelets, which are expressed as $\Psi_{(\boldsymbol{\lambda},z)}, z = 1, \ldots, Z$. By changing the response intensity parameter $\boldsymbol{\lambda}$, one can obtain a diversity of 3-D flexible Gabor filters, which means more spatial–spectral information. It is clear that the chosen parameter $\boldsymbol{\lambda}$ can be irregular, and as much as possible, nevertheless, only five specific selections (including the circular form) are designed and reserved to make a meaningful tradeoff between model representation and complexity. Concretely, $\boldsymbol{\lambda}$ belongs to a predefined set $\boldsymbol{\Omega}$ that includes

$$\boldsymbol{\Omega} = [(1, 1, 1, 1), (2, 2, 2, 2), (1, 2, 2, 5/3), (2, 1, 2, 5/3), (2, 2, 1, 5/3)]. \tag{11}$$

Notice that λ_4 can be computed by the other three parameters [referred to (9)]. Exactly, the visualization of designed 3-D flexible Gabor filters is shown in Fig. 2. Here, the scale and orientation of three scenarios are set as the same values for clarity, that is, $(f, \theta, \varphi) = (1/16, \pi/2, \pi/2)$. More specifically, Fig. 2(a) with $\boldsymbol{\lambda} = (1, 1, 1, 1)$ displays the conventional 3-D Gabor filter, which is isotropic in each direction. Alternatively, Fig. 2(b) with $\boldsymbol{\lambda} = (2, 2, 2, 2)$ and Fig. 2(c) with $\boldsymbol{\lambda} = (1, 2, 2, 5/3)$ show the selected 3-D flexible Gabor filters that can capture the inhomogeneous 3-D spatial–spectral distribution inside hyperspectral image, and thus, they are important complementary descriptors for the circular one. The other

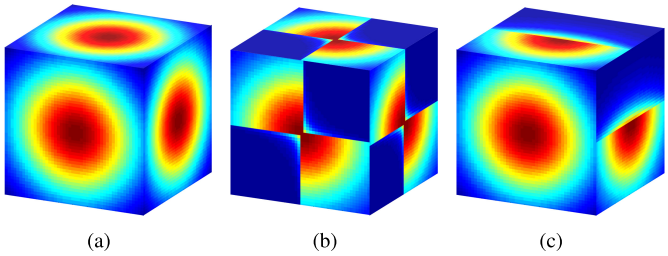


Fig. 2. 3-D flexible Gabor filters with different values of $\boldsymbol{\lambda}$ and the same value of $(f, \theta, \varphi) = (1/16, \pi/2, \pi/2)$. (a) Circular 3-D Gabor filter with $\boldsymbol{\lambda} = (1, 1, 1, 1)$, and designed 3-D flexible Gabor filters with (b) $\boldsymbol{\lambda} = (2, 2, 2, 2)$, and (c) $\boldsymbol{\lambda} = (1, 2, 2, 5/3)$.

two 3-D flexible Gabor filters, i.e., $\boldsymbol{\lambda} = (2, 1, 2, 5/3)$ and $\boldsymbol{\lambda} = (2, 2, 1, 5/3)$, are also taken into account to ensure the integrality of data representation. It is worth to mention that the visualization of the last two flexible Gabor filters is not provided due to the similarity with Fig. 2(c).

Concerning hyperspectral image data cube $\mathbf{R} \in \mathbb{R}^{X \times Y \times B}$ that contains $X \times Y$ spatial pixels and B spectral bands, the 3-D flexible Gabor (FG) features can be computed by

$$\mathbf{G}_{(\boldsymbol{\lambda},z)} = |\Psi_{(\boldsymbol{\lambda},z)} \otimes \mathbf{R}|, \quad \boldsymbol{\lambda} \in \boldsymbol{\Omega}, \ z = 1, \ldots, Z \tag{12}$$

where $\otimes$ is the convolutional operator, and $|\cdot|$ denotes the magnitude of the convolutional results. Several critical issues can be observed from (12). First, the convolutional computation is directly applied on $\mathbf{R}$ without data stretching; hence, the internal spatial–spectral correlation can be fully characterized, and the dimensions of every achieved 3-D FG feature cube are the same as the raw hyperspectral image (RAW), i.e., $\mathbf{G}_{(\boldsymbol{\lambda},z)} \in \mathbb{R}^{X \times Y \times B}$. Second, through varying parameter $\boldsymbol{\lambda}$ with each circumstance of $z, z = 1, \ldots, Z$ (recall that Z equals to 52), 260 modules in total that adequately capture the discriminative representation of materials from different aspects can be, thus, collected. Third, the achieved a large amount of 3-D FG feature cubes indicates a massive computation requirement and may also cause an unstable classification due to the aforementioned Hughes phenomenon. Therefore, an unsupervised LDA-based feature reduction procedure with the aid of superpixel segmentation is applied. For convenience of the following description, the obtained 3-D FG feature cubes are denoted as $\mathbf{G}_{m,z}, m = 1, \ldots, 5; z = 1, \ldots, 52$.

B. Superpixel-Level Unsupervised LDA

In order to provide pseudolabel for our *Su*ULDA algorithm to accomplish the feature reduction task, the simple but effective ERS algorithm presented in Section II-B is introduced. First, a 2-D superpixel map $\mathbf{E} = \{\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_K\} \in \mathbb{R}^{X \times Y}$ with K superpixels is extracted from hyperspectral image $\mathbf{R}$ by utilizing the ERS method, where K can be automatically computed by (6), and $\mathbf{E}_k$ represents the kth superpixel (suppose that N_k pixels are contained within the superpixel), that is

$$\mathbf{E}_k = \left\{\mathbf{E}_{(x,y)} | \mathbf{E}_{(x,y)} = k\right\}, \quad k = 1, \ldots, K. \tag{13}$$

Since the superpixel map $\mathbf{E}$ could well delineate the spatial homogeneity of various materials, it is reasonable to consider each superpixel as a single entity. With respect to the proposed

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*Su*ULDA algorithm, through incorporating the superpixel-level contextual information, the two scatter matrices are constructed from a superpixel viewpoint rather than training samples. In such a situation, the superpixel mechanism is employed to instruct the feature reduction procedure, and the spatial structural representation can be properly reserved. Moreover, instead of applying the *Su*ULDA method on all extracted FG features with a single execution, the feature number of each FG cube is separately reduced.

Second, concerning any of those FG feature cubes, which is denoted as $\mathbf{G}'$ for simplicity, let $\boldsymbol{g}_k \in \mathbb{R}^{N_k \times B}$ be the feature matrix of the kth superpixel $\mathbf{E}_k$ $(k = 1, \ldots, K)$, in which $\boldsymbol{g}_k^i \in \mathbb{R}^B$ is the ith pixel $(i = 1, \ldots, N_k)$. Namely, $\boldsymbol{g}_k^i$ is a feature vector from $\mathbf{G}'$. The mean vector $\boldsymbol{\mu}_k$ of superpixel $\mathbf{E}_k$ is computed as

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} \boldsymbol{g}_k^i. \tag{14}$$

Alternatively, let $\boldsymbol{\mu} \in \mathbb{R}^B$ denote the spatial average vector of the whole FG feature cube $\mathbf{G}'$, which is defined as

$$\boldsymbol{\mu} = \frac{1}{X \times Y} \sum_{k=1}^{K} \sum_{i=1}^{N_k} \boldsymbol{g}_k^i. \tag{15}$$

Thereby, two superpixel-level scatter matrices (i.e., $\mathbf{S}_w$ and $\mathbf{S}_b$) can be derived. More precisely, the within-superpixel scatter matrix $\mathbf{S}_w \in \mathbb{R}^{B \times B}$ depicts the distance between pixels in the same superpixel and the related mean vector, which is calculated by

$$\boldsymbol{S}_w = \sum_{k=1}^{K} \left(\sum_{i=1}^{N_k} \left(\boldsymbol{g}_k^i - \boldsymbol{\mu}_k\right)\left(\boldsymbol{g}_k^i - \boldsymbol{\mu}_k\right)^T \right). \tag{16}$$

Likewise, the between-superpixel scatter matrix $\boldsymbol{S}_b \in \mathbb{R}^{B \times B}$ describes the distance between mean vectors of different superpixels, and it can be computed by

$$\boldsymbol{S}_b = \sum_{k=1}^{K} N_k \left(\boldsymbol{\mu}_k - \boldsymbol{\mu}\right)\left(\boldsymbol{\mu}_k - \boldsymbol{\mu}\right)^T. \tag{17}$$

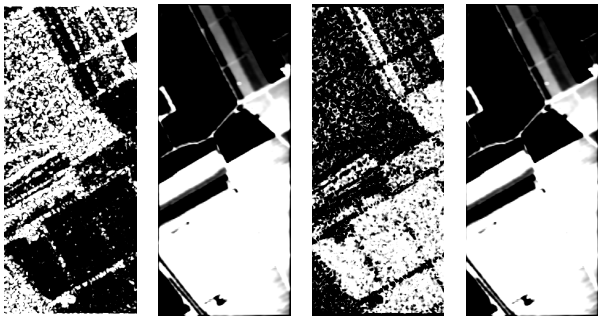
In order to preserve the information of superpixel for classification, the optimization function of our *Su*ULDA at the project direction $\boldsymbol{a}$ can be defined as follows [which is similar to that in (7)]:

$$\arg\max_{\boldsymbol{a}} \frac{\boldsymbol{a}^T \boldsymbol{S}_b \boldsymbol{a}}{\boldsymbol{a}^T \boldsymbol{S}_w \boldsymbol{a}} \tag{18}$$

where $\boldsymbol{a}^T \boldsymbol{S}_b \boldsymbol{a}$ describes the distance between the centers (the center is the mean vector of features after feature reduction in the same superpixel) of different superpixels in subspace. Meanwhile, $\boldsymbol{a}^T \boldsymbol{S}_w \boldsymbol{a}$ represents the covariance of each superpixel in subspace and describes the distance between the pixels in the same superpixel. For optimal discriminative ability, the value of $\boldsymbol{a}^T \boldsymbol{S}_b \boldsymbol{a}$ should be maximized, while the value of $\boldsymbol{a}^T \boldsymbol{S}_w \boldsymbol{a}$ should be minimized.

Similar to the traditional LDA method, the objective function of *Su*ULDA can be solved by

$$\left[\boldsymbol{\gamma}, \mathbf{A}\right] = \mathrm{SVD}\left(\mathbf{S}_w^{-1} \mathbf{S}_b\right) \tag{19}$$



(a) (b) (c) (d)

Fig. 3. Feature maps of FG-*Su*ULDA and FG-PCA on the Salinas hyperspectral image. (a) First channel of 3-D FG-*Su*ULDA feature cube with the values $\boldsymbol{\lambda} = (2, 2, 2, 2)$ and $(f, \theta, \varphi) = (1/2, \pi/4, \pi/2)$. (b) First channel of 3-D FG-PCA feature cube with the same paramters as (a). (c) First channel of 3-D FG-*Su*ULDA feature cube with the values $\boldsymbol{\lambda} = (1, 2, 2, 5/3)$ and $(f, \theta, \varphi) = (1/2, 3\pi/4, \pi/2)$. (d) First channel of 3-D FG-PCA feature cube with the same parameters as (c).

where SVD and $(\cdot)^{-1}$, respectively, denote the singular value decomposition and matrix inversion operators. $\boldsymbol{\gamma} \in \mathbb{R}^B$ is the eigenvalue vector that is sorted from high to low, and $\mathbf{A} \in \mathbb{R}^{B \times B}$ is the corresponding eigenvector matrix. Through picking out d largest eigenvalues from $\boldsymbol{\gamma}$ (it is worth to point out that the value of d is constraint by that the sum of d largest eigenvalues is larger than 90% of the sum of all eigenvalues, and thus, more than 90% energy in 3-D FG feature cube is preserved by d eigenvectors), the reduced eigenvector matrix $\mathbf{A}_d \in \mathbb{R}^{B \times d}$ is utilized for DR of $\mathbf{G}'$, and the compressed feature cube $\mathbf{H}' \in \mathbb{R}^{X \times Y \times d}$ is computed as

$$\mathbf{H}' = \mathbf{G}' \times \mathbf{A}_d. \tag{20}$$

Finally, by employing the *Su*ULDA algorithm on each 3-D FG feature cube $\mathbf{G}_{m,z}, m = 1, \ldots, 5; z = 1, \ldots, 52$, the corresponding FG-*Su*ULDA feature cubes $\mathbf{H}_{m,z}, m = 1, \ldots, 5; z = 1, \ldots, 52$ can be achieved. To be more specific, the feature maps of our *Su*ULDA algorithm on certain FG cubes are displayed in Fig. 3, and the Salinas hyperspectral image is taken into account. Here, the popular unsupervised dimension reduction method, i.e., principal components analysis (PCA), is employed on the FG cubes for comparison (it is worth pointing out that the reduced dimension of PCA method on each FG cube is the same as that of *Su*ULDA). Two circumstances with different FG cubes ($\boldsymbol{\lambda} = (2, 2, 2, 2)$ and $(1, 2, 2, 5/3)$) and orientational parameters ($(f, \theta, \varphi) = (1/2, \pi/4, \pi/2)$ and $(1/2, 3\pi/4, \pi/2)$) are concerned. Compared to the feature maps obtained by FG-PCA [see Fig. 3(b) and (d)] that show a clear outline, the feature maps of the FG-*Su*ULDA algorithm [see Fig. 3(a) and (c)] characterize the contextual information more effectively, which is attributed to the incorporation of superpixel segmentation.

Two advantages can be found from the *Su*ULDA approach. On the one hand, the feature reduction procedure does not need any knowledge of the training set, and the unsupervised property can guarantee the robustness of our method. On the other hand, the discriminative ability has been significantly improved due to the integration of superpixel-level structural information. Besides, since the *Su*ULDA

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method is successively applied to each FG feature module, the computational efficiency of *Su*ULDA can be ensured.

C. Classification

After all the FG-*Su*ULDA feature cubes $\mathbf{H}_{m,z}$, $m = 1,\ldots,5$; $z = 1,\ldots,52$ have been extracted by the proposed FG-*Su*ULDA framework, they are simply concatenated together for classification. Since the dimension of each $\mathbf{H}_{m,z}$ is much smaller than that of $\mathbf{G}_{m,z}$, i.e., $d \ll B$, the efficiency of the following classification procedure can be promoted. Here, the popular SVM method is applied. To implement classification, the one-against-all scheme is utilized for multiclass manipulation, and tenfold cross-validation is used for the definition of SVM parameters. Besides, the linear kernel is adopted to achieve an efficient and accurate classification result.

Algorithm 1 FG-*Su*ULDA Algorithm for Hyperspectral Image Classification

Input: Hyperspectral image $\mathbf{R} \in \mathbb{R}^{X\times Y\times B}$, the training set.
Output: The predicted labels of the test set.

BEGIN
Apply the ERS method on $\mathbf{R}$ to create the superpixel map $\mathbf{E}$;
Use (10) to construct the 3-D flexible Gabor filters $\Psi_{(\lambda,z)}, z = 1,\ldots,52$;
for $m = 1$ to 5 **do**
  for $z = 1$ to 52 **do**
    Use (12) to obtain the 3-D FG feauture cube $\mathbf{G}_{m,z}$;
    Compute the within-superpixel scatter matrix $\boldsymbol{S}_w$ and between-superpixel scatter matrix $\boldsymbol{S}_b$ by (16) and (17), respectively;
    Apply (19) and (20) to calculate the compressed feature cube $\mathbf{H}_{m,z}$;
  end for
end for
Concatenate all the FG-*Su*ULDA feature cubes $\mathbf{H}_{m,z}, m = 1,\ldots,5; z = 1,\ldots,52$, and $\mathbf{H} \in \mathbb{R}^{X\times Y\times 52d}$;
Use the training set from $\mathbf{H}$ to train the SVM model, which is then applied to classify the test set.
End

Finally, the main steps of the proposed FG-*Su*ULDA approach for hyperspectral image classification are summarized in Algorithm 1. It can be clearly observed from Algorithm 1 that the feature extraction (3-D flexible Gabor filtering) and DR (*Su*ULDA scheme) part is unrelated with the training set, which can be computed in a single execution and prepared in advance. Concretely, the computational load of 3-D flexible Gabor filtering is $O(XYB)$, while that of the *Su*ULDA scheme is $O(B^3d)$. Alternatively, the SVM classification part is closely related to the training set, and the computational complexity of SVM is the third power of the number of training samples. It is obvious that the first part has linear computational complexity with the spatial coverage of the hyperspectral image, while the second part is also computational efficient since the small sample set problem is mainly concerned in this article. Thereby, our FG-*Su*ULDA approach has the merit of high calculation efficiency and is suitable for the hyperspectral image with large spatial size.

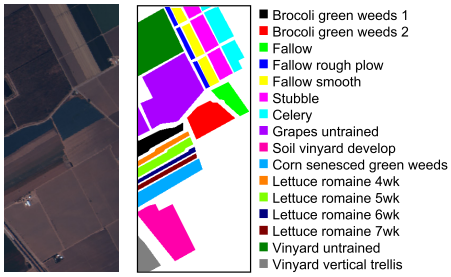


Fig. 4. (a) Pseudocolor illustration and (b) ground-truth map of the Salinas hyperspectral image (16 land-cover classes).

With respect to the parameters contained in our FG-*Su*ULDA approach, they can be either preset (such as the convolutional coefficients in FG model) or determined during algorithm execution (such as the number of superpixels K and compressed feature number d). Therefore, the robustness and generalization of FG-*Su*ULDA can be ensured (the parameters of SVM classifier can be learned by cross-validation).

IV. Experimental Data and Ablation Analysis

In this section, four real hyperspectral image data sets with diverse spatial resolutions in different areas are used to evaluate the module effectiveness in the proposed FG-*Su*ULDA framework. First, the four hyperspectral image data sets are introduced. Second, an ablation experiment is carried out to exploit the contribution of five different sinusoidal waves in formula (11) to the classification results. Third, three designed methods are employed in comparison to show the necessity of FG and *Su*ULDA modules. The ablation experimental results fully demonstrate the effectiveness and rationality of the proposed FG-*Su*ULDA approach.

A. Experimental Data Sets

1) Salinas Hyperspectral Image: The first real data set for testing is the Salinas hyperspectral image that is collected by the airborne visible-infrared imaging spectrometer (AVIRIS) sensor over Salinas Valley, CA, USA. The coverage area comprises 512 lines by 217 samples with 224 spectral bands, and the spatial resolution is 3.7 m per pixel. Only 204 bands are retained after removing 20 water absorption bands. Within the image, 54 129 pixels are labeled, and 16 land-cover classes are provided. The ground-truth map of all labeled samples is shown in Fig. 4, and the detailed information of 16 land covers is described in Table II.

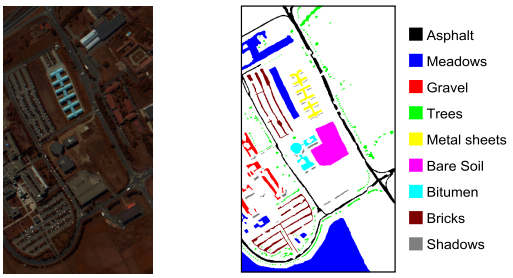
2) Pavia University Hyperspectral Image: The second real hyperspectral image data set is the University of Pavia image, which was collected by the ROSIS-03 sensor over an urban area around the University of Pavia, Pavia, Italy; 103 spectral bands are kept for analysis after omitting 12 noisy bands from the original 115 bands. The spatial size is 610×340, and the spatial resolution is 1.3 m per pixel. The data set contains 42776 labeled samples that are categorized into nine

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TABLE II

LAND-COVER CLASSES WITH THE NUMBER OF LABELED SAMPLES FOR THE SALINAS HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of Samples
$C1$	Brocoli green weeds 1	2009
$C2$	Brocoli green weeds 2	3726
$C3$	Fallow	1976
$C4$	Fallow rough plow	1394
$C5$	Fallow smooth	2678
$C6$	Stubble	3959
$C7$	Celery	3579
$C8$	Grapes untrained	11271
$C9$	Soil vinyard develop	6203
$C10$	Corn senesced green weeds	3278
$C11$	Lettuce romaine 4wk	1068
$C12$	Lettuce romaine 5wk	1927
$C13$	Lettuce romaine 6wk	916
$C14$	Lettuce romaine 7wk	1070
$C15$	Vinyard untrained	7268
$C16$	Vinyard vertical trellis	1807
	Total	54129



(a) (b)

Fig. 5. (a) Pseudocolor illustration and (b) ground-truth map of the Pavia University hyperspectral image (nine land-cover classes).

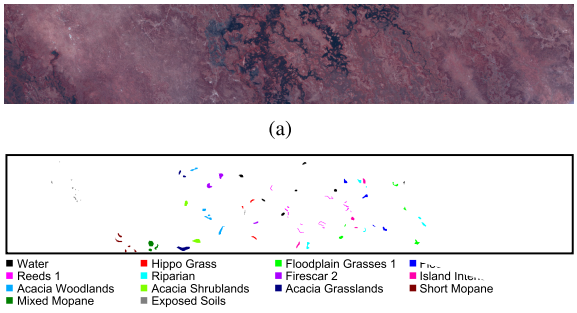
TABLE III

LAND-COVER CLASSES WITH THE NUMBER OF LABELED SAMPLES FOR THE PAVIA UNIVERSITY HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of Samples
C1	Asphalt	6631
C2	Meadows	18649
C3	Gravel	2099
C4	Trees	3064
C5	Painted metal sheets	1345
C6	Bare Siol	5029
C7	Bitumen	1330
C8	Self-Blocking Bricks	3682
C9	Shadows	947
	Tatal	42776

classes of surface materials. The spatial distribution of the Pavia University hyperspectral image is shown in Fig. 5, and the detailed information about labeled samples is listed in Table III.

3) Botswana Hyperspectral Image: The third real hyperspectral data set "Botswana" is acquired by the NASA EO-1 satellite over the Okavango Delta, Botswana, in 2001–2004. The spatial resolution of data is 30 m per pixel. The spatial and spectral dimensions are 1476×256 and 242. After removing the uncalibrated and noisy bands associated with water absorption features, the remaining 145 bands (10–55, 82–97, 102–119, 134–164, and 187–220) are reserved for analysis.



(a) (b)

Fig. 6. (a) Pseudocolor illustration and (b) ground-truth map of the Botswana hyperspectral image (14 land-cover classes).

TABLE IV

LAND-COVER CLASSES WITH THE NUMBER OF LABELED SAMPLES FOR THE BOTSWANA HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of Samples
$C1$	Water	270
$C2$	Hippo Grass	101
$C3$	Floodplain Grasses 1	251
$C4$	Floodplain Grasses 2	215
$C5$	Reeds 1	269
$C6$	Riparian	269
$C7$	Firescar 2	259
$C8$	Island Interior	203
$C9$	Acacia Woodlands	314
$C10$	Acacia Shrublands	248
$C11$	Acacia Grasslands	305
$C12$	Short Mopane	181
$C13$	Mixed Mopane	268
$C14$	Exposed Soils	95
	Total	3248

The data set contains 14 ground-truth classes and 3248 labeled samples. The detailed information is displayed in Fig. 6 and Table IV. It can be seen from Fig. 6 that the materials scatter in the spatial domain, which is a big challenge for our superpixel-oriented methods.

4) Indian Pines Hyperspectral Image: The last real hyperspectral data set is the Indian Pines image, which was gathered by the AVIRIS sensor from Northwestern Indiana. After removing the bands covering the water absorption region, the effective spectral bands' number of the image is 200, and the spatial dimension is 145×145. There are 10 249 labeled pixels (nearly half) in the Indian Pines data set, which are grouped into 16 classes. The spatial resolution is 20 m per pixel with a wavelength range from 0.4 to 2.5μm. The specific spatial information and sample distribution are shown in Fig. 7 and Table V.

B. Contribution of Five Sinusoidal Waves

The classification performance is quantitatively assessed by the overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ), which, respectively, represents the ratio between the correctly classified samples and all test samples, the local correctness, and statistical-oriented classification performance [58]. The higher the values of the three metrics, the greater the discriminative ability and quality of the extracted feature. Besides, due to the aforementioned

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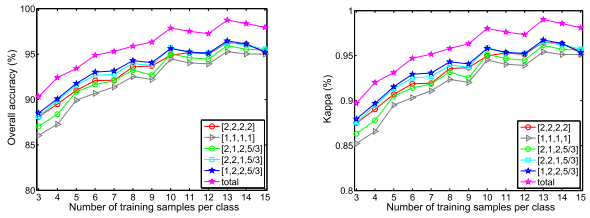


(a) (b)

Fig. 7. (a) Pseudocolor illustration and (b) ground-truth map of the Indian Pines hyperspectral image (16 land-cover classes).

TABLE V
LAND-COVER CLASSES WITH THE NUMBER OF LABELED SAMPLES FOR THE INDIAN PINES HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of Samples
$C1$	Alfalfa	46
$C2$	Corn-notill	1428
$C3$	Corn-mintill	830
$C4$	Corn	237
$C5$	Grass-pasture	483
$C6$	Grass-trees	730
$C7$	Grass-pasture-mowed	28
$C8$	Hay-windrowed	478
$C9$	Oats	20
$C10$	Soybean-notill	972
$C11$	Soybean-mintill	2455
$C12$	Soybean-clean	593
$C13$	Wheat	205
$C14$	Woods	1265
$C15$	Buildings-Grass-Trees-Drives	386
$C16$	Stone-Steel-Towers	93
	Total	10249

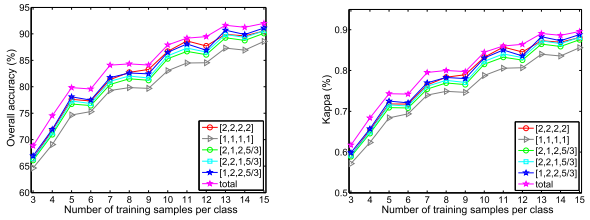


(a) (b)

Fig. 8. Salinas data set. (a) OA and (b) kappa of five sinusoidal waves, respectively, and our flexible Gabor wavelet.

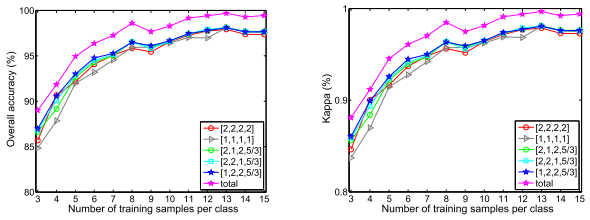
small sample size problem, the training samples are randomly selected with a number of 3–15 for each class and the retained labeled samples form a testing set. Each experiment is repeated ten times with different training samples to ensure reliable results, and the mean values of three quantitative metrics are reported.

In order to investigate the effect of a different sinusoidal wave on the classification performance, a series of experiments are carried out on the four hyperspectral images, as illustrated in Figs. 8–11. Here, the five designed sinusoidal wave is examined, respectively. It can be found in Figs. 8–11 that various sinusoidal wave exhibits different performance on different hyperspectral image data set. For example, with respect to the Salinas data set (as shown in Fig. 8), the filters with a parameter value of (1, 2, 2, 5/3) (dark blue) and (2, 2, 1, 5/3) (sky blue) make the highest contribution to the fusion result. Alternatively, for the Indian Pines image in Fig. 11,



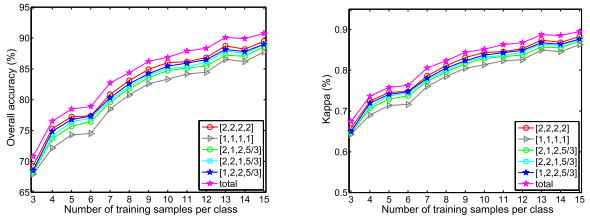
(a) (b)

Fig. 9. Pavia University data set. (a) OA and (b) kappa of five sinusoidal waves, respectively, and our flexible Gabor wavelet.



(a) (b)

Fig. 10. Botswana data set. (a) OA and (b) kappa of five sinusoidal waves, respectively, and our flexible Gabor wavelet.



(a) (b)

Fig. 11. Indian Pines data set. (a) OA and (b) kappa of five sinusoidal waves, respectively, and our flexible Gabor wavelet.

the sinusoidal wave with parameter value (2, 2, 2, 2) (red) has the highest contribution to the classified result. Therefore, since the contribution of different sinusoidal waves varies in the classification results of diverse hyperspectral images, it is necessary to consider the five different sinusoidal waves in a unified model, and thus, the stability and universality of the proposed FG-*Su*ULDA can be ensured.

C. Module Ablation Analysis

In order to demonstrate the effectiveness of the two modules, both FG and *Su*ULDA, three methods are taken into consideration. Specifically, the 3-D circular Gabor magnitude (CGabor) feature only makes use of the circular Gabor wavelets [referred to (1)], and all the obtained circular Gabor features are directly concatenated together for classification without any feature reduction operation. Meanwhile, the CGabor feature is combined with the designed *Su*ULDA scheme to form the so-called CG-*Su*ULDA. This comparison will demonstrate the effectiveness of the proposed 3-D FG feature. It is worth to point out that the FG features without *Su*ULDA-based DR are not conducted in comparison due to infeasible computational requirements. Besides, as illustrated in Fig. 3, the FG-PCA technique is also concerned

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TABLE VI

CLASSIFICATION PERFORMANCE OF THE FOUR ABLATIVE APPROACHES ON THE SALINAS HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	CGabor	CG-*Su*ULDA	FG-PCA	FG-*Su*ULDA
C1	59.10	69.67	72.20	**92.95**
C2	77.75	93.99	76.03	**99.86**
C3	58.14	92.62	70.44	**100.00**
C4	71.52	74.93	91.34	**98.74**
C5	72.37	85.60	83.08	**95.29**
C6	66.73	92.35	94.30	**99.83**
C7	63.64	91.17	93.76	**96.95**
C8	38.83	65.01	69.38	**71.22**
C9	69.19	98.13	94.81	**99.99**
C10	52.42	71.47	69.56	**88.82**
C11	88.40	98.67	92.54	**99.93**
C12	65.33	74.31	84.02	**98.75**
C13	72.48	78.96	90.33	**94.73**
C14	73.95	66.92	89.73	**90.25**
C15	48.42	83.92	57.61	**91.26**
C16	75.19	**99.90**	84.58	99.86
OA	59.13	82.35	78.24	**91.07**
AA	65.84	83.60	82.11	**94.90**
kappa	0.55	0.80	0.76	**0.90**

TABLE VII

CLASSIFICATION PERFORMANCE OF THE FOUR ABLATIVE APPROACHES ON THE PAVIA UNIVERSITY HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	CGabor	CG-*Su*ULDA	FG-PCA	FG-*Su*ULDA
C1	35.62	39.54	49.19	**50.74**
C2	60.51	64.69	63.58	**67.61**
C3	55.52	70.40	63.56	**86.07**
C4	**74.99**	54.80	70.32	56.45
C5	95.12	93.97	98.06	**99.59**
C6	46.23	77.91	55.32	**94.87**
C7	56.61	74.86	75.27	**99.25**
C8	35.44	55.46	63.30	**64.16**
C9	35.94	37.20	**93.39**	42.26
OA	54.03	61.75	62.94	**69.44**
AA	55.11	63.20	70.22	**73.45**
kappa	0.43	0.53	0.54	**0.62**

TABLE VIII

CLASSIFICATION PERFORMANCE OF THE FOUR ABLATIVE APPROACHES ON THE BOTSWANA HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	CGabor	CG-*Su*ULDA	FG-PCA	FG-*Su*ULDA
C1	84.68	67.64	**95.81**	80.34
C2	73.67	88.27	95.31	**97.65**
C3	76.33	68.06	82.06	**92.14**
C4	77.55	77.26	69.58	**85.61**
C5	48.12	68.05	59.89	**79.92**
C6	58.05	59.06	57.71	**78.42**
C7	93.52	92.30	90.16	**96.60**
C8	79.00	71.65	**91.60**	80.90
C9	77.40	94.31	77.17	**98.33**
C10	**98.69**	93.55	95.92	86.04
C11	64.64	81.92	78.18	**83.64**
C12	65.51	**93.15**	73.20	91.24
C13	51.21	99.02	66.91	**100.00**
C14	84.57	75.22	84.78	**86.09**
OA	72.98	80.93	78.76	**88.16**
AA	72.95	81.10	79.97	**88.53**
kappa	0.70	0.79	0.77	**0.87**

TABLE IX

CLASSIFICATION PERFORMANCE OF THE FOUR ABLATIVE APPROACHES ON THE INDIAN PINES HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	CGabor	CG-*Su*ULDA	FG-PCA	FG-*Su*ULDA
C1	93.70	88.37	82.17	**98.26**
C2	87.90	92.90	71.50	**97.96**
C3	45.29	43.51	41.66	**52.23**
C4	54.88	64.34	49.97	**68.94**
C5	**99.61**	98.82	80.59	96.86
C6	46.66	56.97	34.32	**66.68**
C7	67.02	70.47	66.76	**72.49**
C8	**84.05**	76.72	81.92	82.87
C9	84.43	83.95	77.90	**84.59**
C10	98.26	95.22	95.22	**100.0**
C11	92.21	**96.49**	79.48	95.93
C12	98.82	91.18	84.12	**100.0**
C13	39.42	43.51	**54.91**	54.88
C14	49.25	59.55	41.78	**65.37**
C15	**79.85**	64.03	59.14	77.11
C16	96.60	87.18	85.98	**98.23**
OA	61.22	63.76	56.59	**70.82**
AA	76.12	75.83	67.96	**82.03**
kappa	0.57	0.60	0.52	**0.67**

in the experiments to exhibit the importance of the proposed *Su*ULDA module. All these methods use the SVM classifier under the same setting for classification.

Tables VI–IX, respectively, show the classification performance of the four compared methods on the four hyperspectral image data sets. Here, only three samples per class are randomly picked out from the labeled samples to constitute the training set, and the retained labeled ones compose the testing set. The procedure is repeated ten times, and the mean values are reported. It is interesting to find from Tables VI–IX that the CGabor technique almost performs the worst in terms of classification accuracy on all the four data sets. This can be explained that although the extracted circular Gabor features contain important discriminative information, which also brings large redundancy, the brutal concatenation of all circular Gabor features could have a negative (rather than positive) effect on the classification performance. More severely, the Hughes phenomenon can be further exacerbated due to the huge number of connected Gabor features, which conversely confirms the importance and necessity of feature reduction procedure.

Generally, the two FG-based methods, both FG-PCA and FG-*Su*ULDA, outperform CGabor, indicating the necessity of the FG module. Alternatively, the two *Su*ULDA-based techniques, both CG-*Su*ULDA and FG-*Su*ULDA, respectively, achieve better results than the compared CGabor and FG-PCA, demonstrating the effectiveness of the developed *Su*ULDA module. In particular, concerning the classification results on the Salinas data set given in Table VI, the newly designed CG-*Su*ULDA is significantly higher than the CGabor technique. This can be ascribed to that the *Su*ULDA scheme could effectively discard redundant features, and the integration of superpixel-level structural information could greatly improve the discriminative ability of the compressed Gabor features.

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Alternatively, with respect to the classification results on the Pavia University data set given in Table VII, the CG-*Su*LDA method remarkably improves the classification results from 54.03%, 55.11%, and 0.43 of CGabor to 61.75%, 63.20%, and 0.53 in the OA, AA, and kappa coefficient metrics, which is mainly due to the incorporation of the superpixel-level unsupervised LDA feature reduction method. Moreover, the proposed FG-*Su*LDA approach utilizes a flexible Gabor form to obtain more discriminative information, achieving the best performance in all contrastive methods. In addition, the classification maps of the four comparison methods on the four hyperspectral image data sets are, respectively, shown in Figs. 13, 15, 17, and 19, and the same conclusion can be drawn.

V. Experimental Results

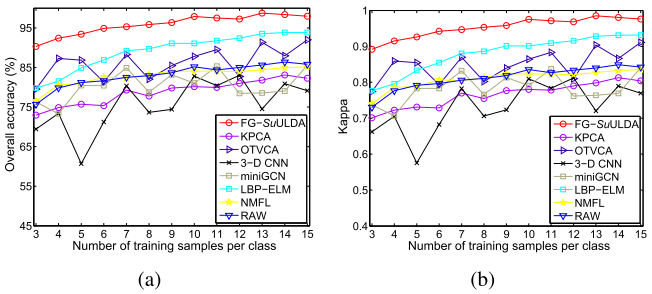
In this section, the experiments are designed for assessing the superiority and rationality of FG-*Su*LDA framework. First, four state-of-the-art spatial–spectral feature extraction frameworks, including nonlinear multiple feature learning-based classification (NMFL) [59], LBPs and extreme learning machine (LBP-ELM) [27], 3-D convolutional neural network (3-D CNN) [60], and a graph convolutional network-based method (miniGCN) [61], are chosen for comparison. Second, three different kinds of features, i.e., RAW, KPCA, and orthogonal total variation component analysis (OTVCA) [62], are compared with the FG-*Su*LDA-extracted feature. In summary, a diversity of combinations of feature extraction methods, i.e., RAW, KPCA, OTVCA, NMFL, LBP-ELM, 3-D CNN, and miniGCN, is investigated in the experiments.

Among all methods in comparison, the four methods (NMFL, LBP-ELM, 3-D CNN, and miniGCN) utilize their built-in classifiers, while others use the SVM classifier under the same setting. Likewise, the classification performance of all compared methods is assessed by OA, AA, and kappa coefficient. The training set is composed of 3–15 points randomly selected from each class, and the three quantitative metrics are obtained from the average values of ten different training scenarios.

Furthermore, the running time (in seconds) of the compared techniques on the five hyperspectral image data sets is depicted in Table X. Here, three samples per class are used for training, and the average of ten experiments is computed. Generally, the computing time is recorded by a workstation with a 24-core Intel processor of 2.20 GHz and 128-GB RAM. Alternatively, the two deep learning-based methods, including 3-D CNN and miniGCN, which usually needs graphics processing unit (GPU) to accelerate the computation, were trained in an environment of a four-core/eight-way multitask Intel Xeon E-5 2699 processor with a processing speed of 2.20 GHz and two Tesla P100 GPUs.

A. Salinas Hyperspectral Image

The experimental results with the aforementioned methods, including RAW, KPCA, OTVCA, NMFL, LBP-ELM, 3-D CNN, miniGCN, and FG-*Su*LDA, are demonstrated in Fig. 12. The number of training samples per class is $(3, 4, \ldots, 15)$. An increasing number of training numbers lead to improved classification performance for all feature extraction frameworks, as demonstrated in Fig. 12. It can be seen that the performance of 3D-CNN is lower than the RAW method due to the overfitting problem of deep learning methods with a very limited training sample set. Three state-of-the-art methods, including OTVCA, NMFL, and LBP-ELM, generally provide higher accuracies than the RAW method. Overall, the proposed FG-*Su*LDA always achieves the best classification results, especially with the small training set, demonstrating its superiority and the power of the introduced flexible Gabor features.



(a) (b)

Fig. 12. Classification accuracy versus the number of training samples per class on the Salinas hyperspectral image. (a) OA. (b) Kappa.

TABLE X
Running Time (in Seconds) for Different Methods When Three Samples Per Class Are Used for Training

Method	Salinas	Pavia University	Botswana	Indian Pines
RAW	1.82	2.89	3.85	1.09
KPCA	46.16	53.01	83.06	7.40
OTVCA	220.82	410.66	781.29	81.26
NMFL	1.952	3.634	11.89	1.277
LBP-ELM	30.73	31.63	43.06	5.50
3-D CNN	197.29	242.63	409.43	101.13
miniGCN	905.4	1166.2	1194.0	310.96
CGabor	89.14	113.26	224.18	28.82
CG-*Su*LDA	106.07	140.89	293.71	40.31
FG-PCA	537.91	888.06	1077.12	162.74
FG-*Su*LDA	395.46	670.25	810.58	127.84

Considering the scarce training sample problem in practice, the accuracies for each class and the three metrics (OA, AA, and kappa coefficient) are listed in Table XI. As seen from the table, the proposed FG-*Su*LDA framework performs better than the other methods and achieves the best accuracies for most classes. The advantage of the proposed FG-*Su*LDA approach can also be verified by the classification maps (they are obtained in a single experiment), as displayed in Fig. 13. It can be obviously observed that the two *Su*LDA-based classification maps [i.e., Fig. 13(j) and (l)] are more regular than the nine alternatives [see Fig. 13(b)–(i), (k)], indicating the necessity of integrating the superpixel-level spatial structural information. Furthermore, the proposed FG-*Su*LDA exhibits better performance in the edge area of classes than CG-*Su*LDA, confirming the indispensability of the employed flexible Gabor operators.

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TABLE XI

CLASSIFICATION PERFORMANCE OF THE EIGHT COMPARED APPROACHES ON THE SALINAS HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	RAW	KPCA	OTVCA	NMFL	LBP-ELM	3-D CNN	miniGCN	FG-*Su*LDA
C1	95.81	92.49	**99.35**	99.21	90.23	89.68	95.41	92.95
C2	89.74	82.88	89.66	98.36	87.88	67.69	82.08	**99.86**
C3	73.57	72.45	83.78	81.83	84.67	63.80	65.38	**100.00**
C4	98.94	99.17	97.56	74.96	94.13	93.74	**99.50**	98.74
C5	87.87	84.66	66.24	91.94	85.78	70.23	64.97	**95.29**
C6	98.38	99.01	99.70	95.93	78.49	90.21	99.37	**99.83**
C7	91.20	95.40	98.71	96.47	79.38	93.67	**99.50**	96.95
C8	49.33	47.71	44.36	47.60	**77.61**	46.60	52.64	71.22
C9	92.37	86.87	98.37	95.56	74.34	94.16	95.08	**99.99**
C10	68.30	62.42	61.34	79.12	76.46	55.09	70.93	**88.82**
C11	74.06	86.74	94.84	93.41	93.62	62.83	90.05	**99.93**
C12	95.26	90.70	**100.0**	85.01	72.34	87.33	85.14	98.75
C13	97.71	72.18	**99.23**	87.83	79.56	88.87	86.97	94.73
C14	88.45	86.15	87.63	76.17	70.61	67.31	86.88	**90.25**
C15	55.34	55.77	88.58	49.23	75.26	56.19	59.75	**91.26**
C16	74.33	55.10	62.25	92.49	92.21	55.22	87.20	**99.86**
OA	75.69	72.93	79.56	76.52	79.80	69.41	76.28	**91.07**
AA	83.17	79.36	85.72	84.07	82.04	73.93	82.55	**94.90**
kappa	0.73	0.70	0.77	0.74	0.78	0.66	0.74	**0.90**

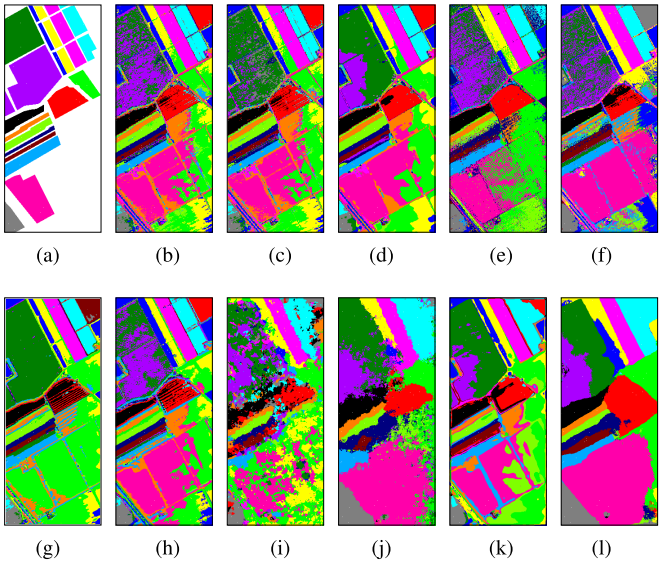
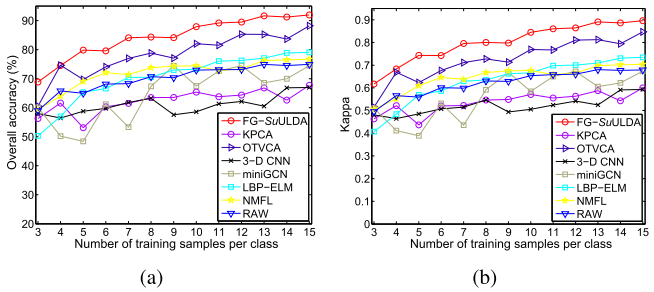


Fig. 13. Salinas hyperspectral image. (a) Testing map (three labeled samples per class for training) and classification maps obtained by (b) RAW (72.28%), (c) KPCA (72.21%), (d) OTVCA (80.03%), (c) NMFL (71.13%), (d) LBP-ELM (83.30%), (e) 3-D CNN (68.98%), (e) miniGCN (77.02%), (f) CGabor (58.32%), (g) CG-*Su*LDA (85.75%), (h) FG-PCA (78.04%), and (i) FG-*Su*LDA (92.08%). The numbers in the brackets are the OA values. (a) Testing map. (b) RAW. (c) KPCA. (d) OTVCA. (e) NMFL. (f) LBP-ELM. (g) 3-D CNN. (h) miniGCN. (i) CGabor. (j) CG-*Su*LDA. (k) FG-PCA. (l) FG-*Su*LDA.

B. Pavia University Hyperspectral Image

Fig. 14 displays the classification results of the FG-*Su*LDA approach and other compared methods with different training samples. Since the spatial distribution of most materials is not as regular as that in the Salinas hyperspectral image, the classification accuracies in all cases are decreased. The two deep learning-based methods, both 3-D CNN and miniGCN, provides the worst results in the classification of scarce training samples. Besides, the KPCA dimension reduction method



(a) (b)

Fig. 14. Classification accuracy versus the number of training samples per class on the Pavia University hyperspectral image. (a) OA. (b) Kappa.

TABLE XII

CLASSIFICATION PERFORMANCE OF THE EIGHT COMPARED APPROACHES ON THE PAVIA UNIVERSITY HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	RAW	KPCA	OTVCA	NMFL	LBP-ELM	3-D CNN	miniGCN	FG-*Su*LDA
C1	53.03	48.45	55.11	**63.98**	37.00	46.18	34.60	50.74
C2	54.66	53.52	57.40	52.50	43.81	58.62	56.09	**67.61**
C3	52.31	43.08	50.70	43.57	59.64	54.59	57.49	**86.07**
C4	76.32	77.89	82.91	76.50	51.24	75.01	**94.18**	56.45
C5	98.19	98.30	96.57	91.98	85.36	84.35	98.21	**99.59**
C6	53.24	42.95	52.89	59.26	70.53	39.20	61.48	**94.87**
C7	77.43	82.84	88.58	86.50	70.84	73.62	88.24	**99.25**
C8	57.34	55.86	50.11	51.76	55.69	52.92	63.66	**64.16**
C9	99.88	**99.95**	99.92	85.84	39.60	95.47	99.89	42.26
OA	58.98	56.27	60.52	59.32	50.27	56.98	60.13	**69.44**
AA	69.16	66.98	70.47	67.99	57.08	64.44	72.65	**73.45**
kappa	0.50	0.46	0.52	0.51	0.41	0.47	0.51	**0.62**

loses a large amount of effective information, which affects the classification accuracy. Nevertheless, due to the flexible Gabor and *Su*LDA modules, the proposed FG-*Su*LDA approach achieves the optimal result among all contrastive methods.

Likewise, when the training set only contains three labeled samples per class, the detailed classification performance of various compared methods is presented in Table XII. The same conclusion can be drawn that the proposed FG-*Su*LDA method gets the highest results in most classes and the three performance measures. The classification maps of all compared methods are displayed in Fig. 15. It can be visually observed that the classification map of our FG-*Su*LDA [see Fig. 15(l)] is in better accordance with the testing map [see Fig. 15(a)].

C. Botswana Hyperspectral Image

The classification results by eight compared methods are illustrated in Fig. 16, with varying sizes of training samples. Likewise, our FG-*Su*LDA provides the best accuracies in all circumstances, indicating the importance of the flexible Gabor extension. Again, when the training set is only three samples per class, the per-class accuracy and classification maps of eight methods in comparison are provided in Table XIII and Fig. 17, respectively. All these results constantly demonstrate the superiority and effectiveness of our FG-*Su*LDA approach even though the spatial distribution of materials in the data is indistinct.

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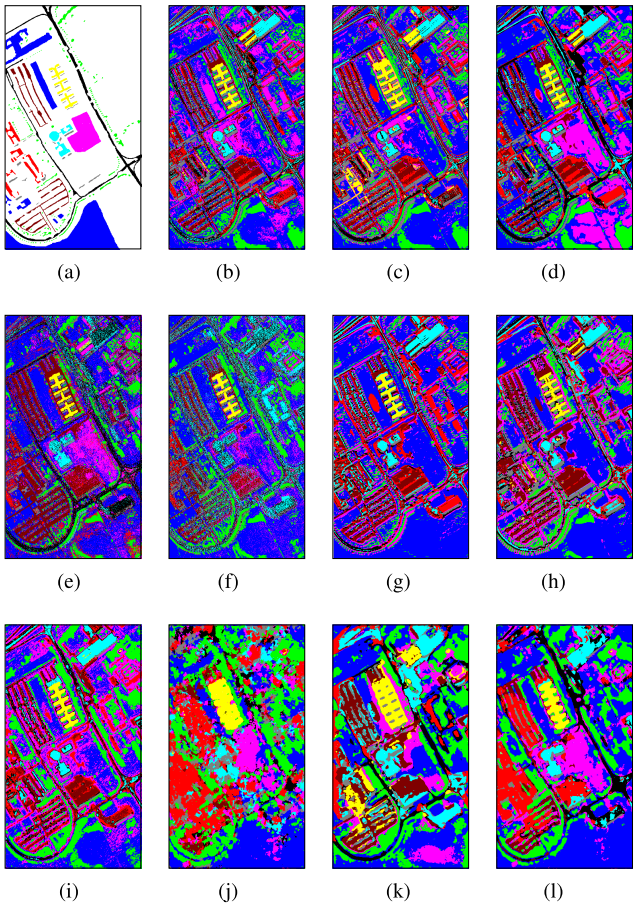


Fig. 15. Pavia University hyperspectral image. (a) Resting map (three labeled samples per class for training) and classification maps obtained by (b) RAW (58.65%), (c) KPCA (56.86%), (d) OTVCA (61.25%), (e) NMFL (57.00%), (f) LBP-ELM (54.49%), (g) 3-D CNN (58.51%), (h) miniGCN (62.05%), (i) CGabor (55.12%), (j) CG-*Su*LDA (68.58%), (k) FG-PCA (61.22%), and (l) FG-*Su*LDA (74.31%). The numbers in the brackets are the OA values. (a) Testing map. (b) RAW. (c) KPCA. (d) OTVCA. (e) NMFL. (f) LBP-ELM. (g) 3-D CNN. (h) miniGCN. (i) CGabor. (j) CG-*Su*LDA. (k) FG-PCA. (l) FG-*Su*LDA.

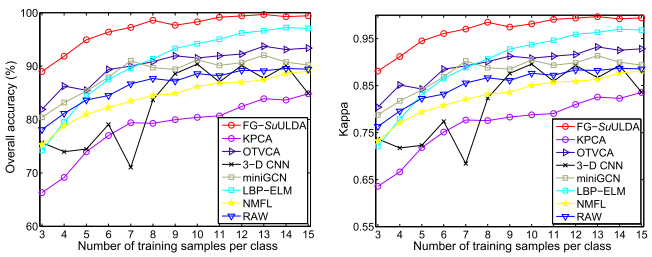


Fig. 16. Classification accuracy versus the number of training samples per class on the Botswana hyperspectral image. (a) OA. (b) Kappa.

D. Indian Pines Hyperspectral Image

The Indian Pines hyperspectral image is always a challenging data for classification due to the low spatial resolution and the number of land covers. Fig. 18 shows the classification results with different training sample numbers by our FG-*Su*LDA and seven contrastive methods.

TABLE XIII

CLASSIFICATION PERFORMANCE OF THE EIGHT COMPARED APPROACHES ON THE BOTSWANA HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	RAW	KPCA	OTVCA	NMFL	LBP-ELM	3-D CNN	miniGCN	FG-*Su*LDA
C1	99.25	99.25	**100.0**	97.38	74.83	99.89	**100.0**	80.34
C2	82.45	82.14	94.13	89.29	79.08	69.90	**98.98**	97.65
C3	85.65	82.26	67.84	85.28	61.77	**92.18**	73.39	92.14
C4	81.27	65.80	81.96	**91.75**	62.22	91.65	66.51	85.61
C5	71.58	58.93	63.16	67.07	65.79	45.75	**80.08**	79.92
C6	53.42	30.36	50.56	67.63	62.37	60.41	47.74	**78.42**
C7	87.54	75.68	**98.54**	88.52	86.29	96.09	98.05	96.60
C8	84.30	72.25	80.50	**88.70**	74.55	72.60	81.50	80.90
C9	69.71	47.99	59.65	82.86	66.85	68.33	93.25	**98.33**
C10	73.51	57.55	**100.0**	46.90	86.04	70.49	71.02	86.04
C11	72.52	71.52	88.74	62.68	79.27	68.71	**96.36**	83.64
C12	94.27	81.18	**100.0**	52.02	75.28	86.52	98.31	91.24
C13	65.58	45.00	91.89	66.87	96.08	69.13	48.30	**100.00**
C14	**99.35**	98.91	96.74	71.74	63.48	69.78	85.87	86.09
OA	78.12	66.31	81.91	75.28	74.19	75.66	80.41	**88.16**
AA	78.54	69.20	83.84	75.92	74.65	75.82	81.38	**88.53**
kappa	0.76	0.64	0.80	0.75	0.74	0.74	0.79	**0.87**

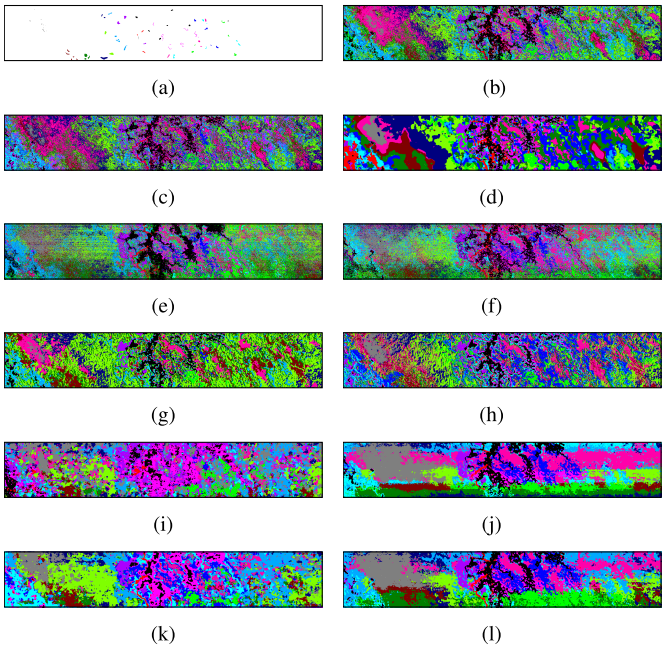
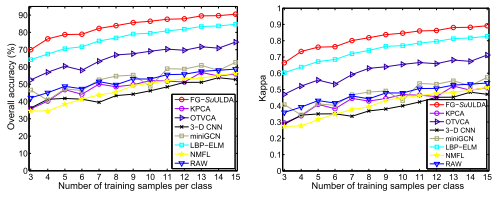


Fig. 17. Botswana hyperspectral image. (a) Testing map (three labeled samples per class for training) and classification maps obtained by (b) RAW (80.29%), (c) KPCA (66.19%), (d) OTVCA (81.19%), (e) NMFL (79.04%), (f) LBP-ELM (78.63%), (g) 3-D CNN (73.14%), (h) miniGCN (80.49%), (i) CGabor (76.29%), (j) CG-*Su*LDA (82.19%), (k) FG-PCA (81.47%), and (l) FG-*Su*LDA (90.80%). The numbers in the brackets are the OA values. (a) Testing map. (b) RAW. (c) KPCA. (d) OTVCA. (e) NMFL. (f) LBP-ELM. (g) 3-D CNN. (h) miniGCN. (i) CGabor. (j) CG-*Su*LDA. (k) FG-PCA. (l) FG-*Su*LDA.

The classification accuracy of NMFL, 3-D CNN, miniGCN, and KPCA approaches is pretty low, while our FG-*Su*LDA achieves the best result, indicating that the proposed FG-*Su*LDA approach has excellent generalization ability with different spatial resolutions. Similarly, Table XIV shows the classification accuracy of eight methods when the number of samples for each class is 3, and the corresponding classification maps with background points are displayed

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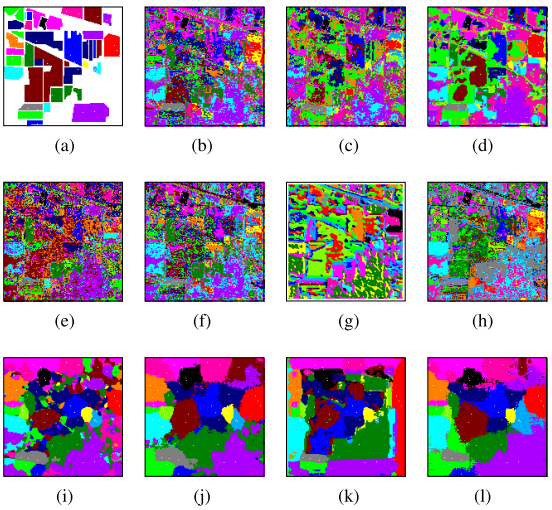
(a) (b)

Fig. 18. Classification accuracy versus the number of training samples per class on the Indian Pine hyperspectral image. (a) OA. (b) Kappa.

TABLE XIV

CLASSIFICATION MEAN OF THE ELEVEN COMPARED APPROACHES ON THE INDIAN PINES HYPERSPECTRAL IMAGE (ONLY THREE LABELED SAMPLES PER CLASS ARE USED TO TRAIN THE CLASSIFIER)

Class	RAW	KPCA	OTVCA	NMFL	LBP-ELM	3-D CNN	miniGCN	FG-*Su*ULDA
C1	89.46	72.55	94.29	71.09	93.48	59.39	85.22	**98.26**
C2	58.72	61.57	72.74	42.10	**99.24**	23.66	38.42	97.96
C3	34.73	44.52	**55.93**	26.94	41.91	20.74	42.55	52.23
C4	41.71	33.26	36.66	34.80	66.34	32.97	32.48	**68.94**
C5	70.20	67.16	81.37	38.24	**99.61**	42.68	40.20	96.86
C6	28.84	30.52	33.92	22.19	51.28	49.02	44.89	**66.68**
C7	64.47	53.24	81.83	54.68	67.27	**84.80**	65.71	72.49
C8	50.50	40.98	67.06	51.80	**86.64**	81.46	49.54	82.87
C9	28.44	24.87	44.76	34.59	82.86	**85.88**	17.24	84.59
C10	89.13	85.87	95.65	85.65	95.65	36.70	88.26	**100.0**
C11	45.58	32.79	51.84	26.75	93.33	39.18	43.68	**95.93**
C12	80.00	70.59	**100.0**	71.18	**100.0**	16.87	74.12	**100.0**
C13	26.06	12.89	24.90	21.30	45.40	**72.18**	21.20	54.88
C14	36.20	35.30	49.58	36.81	52.26	49.51	50.12	**65.37**
C15	56.08	28.39	70.83	63.56	74.44	13.13	**91.84**	77.11
C16	82.97	72.85	97.73	78.18	92.92	45.61	97.13	**98.23**
OA	42.10	35.76	52.48	38.73	64.20	37.97	46.67	**70.82**
AA	55.19	47.96	66.19	47.49	77.66	47.11	55.16	**82.03**
kappa	0.36	0.29	0.47	0.32	0.60	0.31	0.41	**0.67**



(a) (b) (c) (d)

(e) (f) (g) (h)

(i) (j) (k) (l)

Fig. 19. Indian Pines hyperspectral image. (a) Testing map (three labeled samples per class for training) and classification maps obtained by (b) RAW (44.92%), (c) KPCA (36.79%), (d) OTVCA (53.26%), (e) NMFL (40.85%), (f) LBP-ELM (64.20%), (g) 3-D CNN (37.48%), (h) miniGCN (46.73%), (i) CGabor (61.33%), (j) CG-*Su*ULDA (65.86%), (k) FG-PCA (55.17%), and (l) FG-*Su*ULDA (71.20%). The numbers in the brackets are the OA values (a) testing map. (b) RAW. (c) KPCA. (d) OTVCA. (e) NMFL. (f) LBP-ELM. (g) 3-D CNN. (h) miniGCN. (i) CGabor. (j) CG-*Su*ULDA. (k) FG-PCA. (l) FG-*Su*ULDA.

in Fig. 19. It can be intuitively seen that the classification map obtained by our FG-*Su*ULDA algorithm is more structured than the compared ones.

VI. CONCLUSION

In this article, we have proposed an FG-*Su*ULDA approach to address the small sample size problem of hyperspectral image classification. On the one hand, different forms of sinusoidal wave are introduced to enrich the 3-D Gabor wavelets, and a series of 3-D flexible Gabor filters are rigorously designed to fully exploit the diverse spatial–spectral characteristics of materials. On the other hand, a superpixel segmentation map (here, the computational-efficient ERS method is utilized) is employed to guide the feature reduction procedure of LDA. With respect to the *Su*ULDA module, the superpixel guidance skillfully converts the traditional supervised LDA method (depending on the training samples) into an unsupervised manner, and the robustness of *Su*ULDA is greatly improved. Meanwhile, through incorporating the superpixel-level spatial structural information, the most discriminative features can be achieved after the *Su*ULDA feature reduction. Our FG-*Su*ULDA framework also has the merits of powerful generalization (due to the heuristical determination of parameters) and high computational efficiency (due to the irrelevance of feature extraction with the training set and the successive manipulation of *Su*ULDA on each FG feature module).

Four real-world hyperspectral image data sets with various spatial and spectral dimensions are taken into consideration, and a number of alternatives, i.e., RAW, KPCA, OTVCA, NMFL, LBP-ELM, 3-D CNN, miniGCN, CGabor, CG-*Su*ULDA, and FG-PCA, are examined for comparison. Numerical and visual experimental results constantly validate the superiority of the proposed FG-*Su*ULDA approach, including the necessity of flexible Gabor filters and the effectiveness of the superpixel-level unsupervised LDA module. It is worth to point out that the *Su*ULDA scheme proposed here is a valuable unsupervised DR technique and can be applied for other applications.

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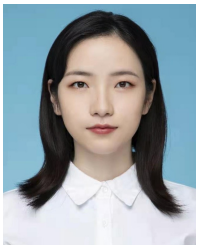
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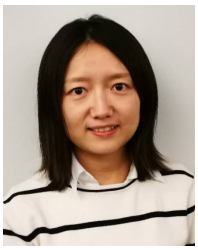
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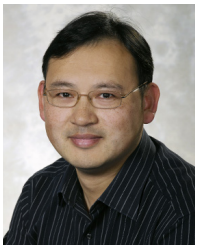
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